DATA COMMUNICATION AND NETWORKING LAB RECORD

Group Number : 14

Roll Number :- IEC2023073

Experiment 4: Static IP Routing in Cisco Packet Tracer

Aim

To design and configure a multi-router network in Cisco Packet Tracer using static routing and verify communication between hosts in different IP networks.

Theory

Static routing is a routing method in which routes are manually configured in every router. Unlike dynamic routing protocols, routers do not exchange route information automatically. Therefore, each router must be told how to reach remote networks. In this experiment three routers were connected in a chain and static routes were added so that communication could occur between all LANs.

Network Topology

The network consists of three routers, three switches and six PCs. Router0 is connected to the left LAN (192.168.1.0/24), Router1 to the middle LAN (192.168.2.0/24) and Router3 to the right LAN (192.168.3.0/24).

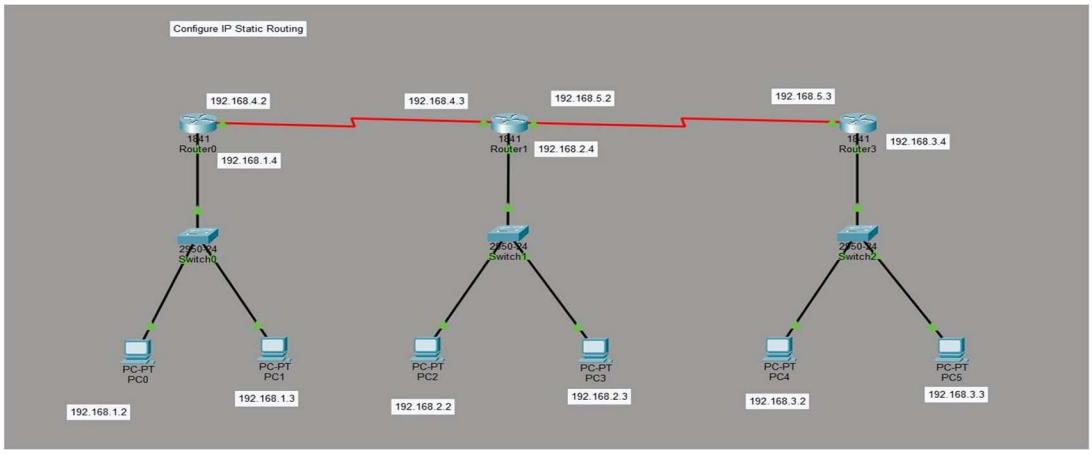

Figure 1 Network Topology

Figure 1 shows the complete topology with the IP addresses of all router interfaces and PCs. The serial link between Router0 and Router1 uses the 192.168.4.0/24 network, while the serial link between Router1 and Router3 uses the 192.168.5.0/24 network.

Static Route Configuration

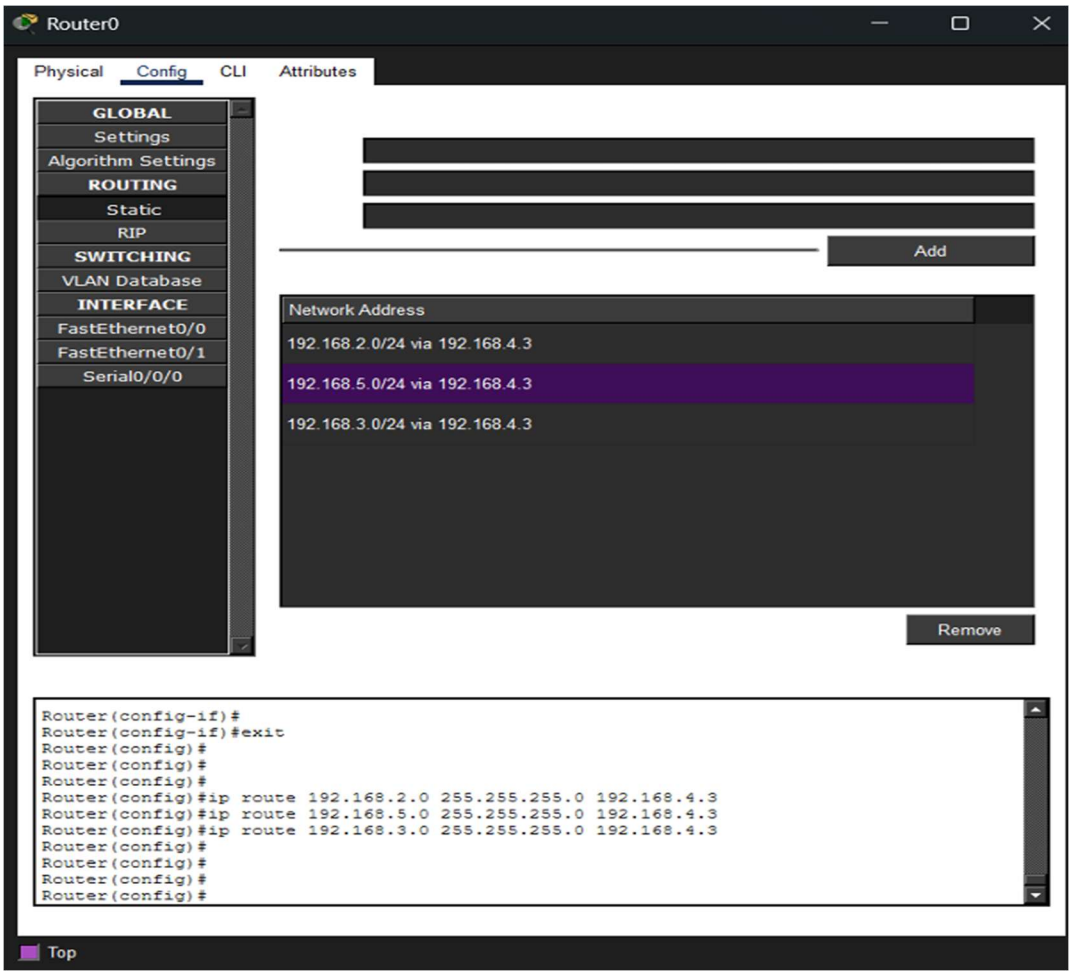

Figure 2 Router0 Routing Table

Figure 2 shows the static routes configured in Router0. Router0 does not know about the middle LAN, right LAN and Router1–Router3 serial network by default. Therefore, the following routes were added:

ip route 192.168.2.0 255.255.255.0 192.168.4.3
ip route 192.168.3.0 255.255.255.0 192.168.4.3
ip route 192.168.5.0 255.255.255.0 192.168.4.3

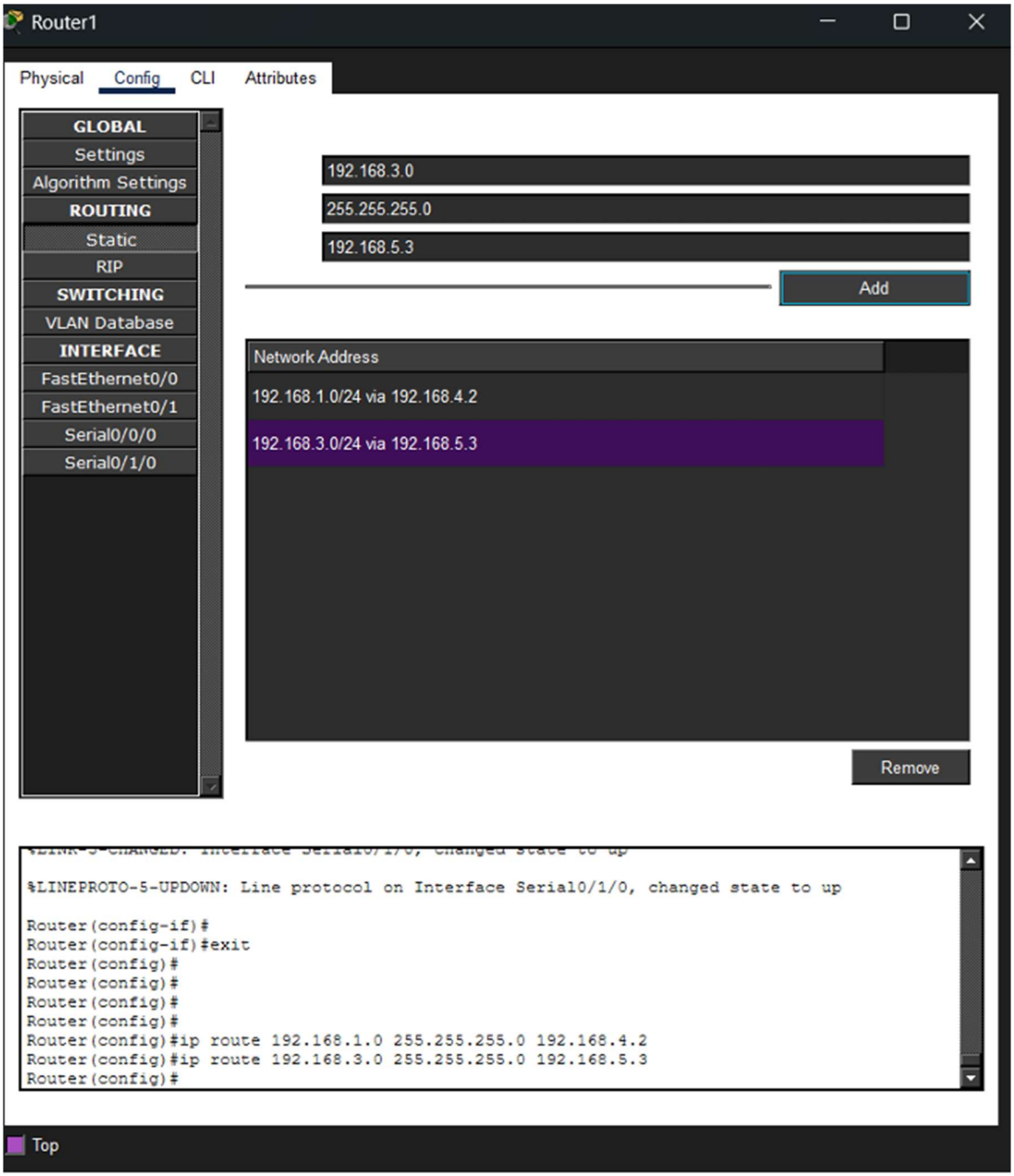

Figure 3 Router1 Routing Table

Figure 3 shows the configuration of Router1. Router1 acts as the central forwarding router. It requires one route to the left LAN through Router0 and another route to the right LAN through Router3.

ip route 192.168.1.0 255.255.255.0 192.168.4.2
ip route 192.168.3.0 255.255.255.0 192.168.5.3

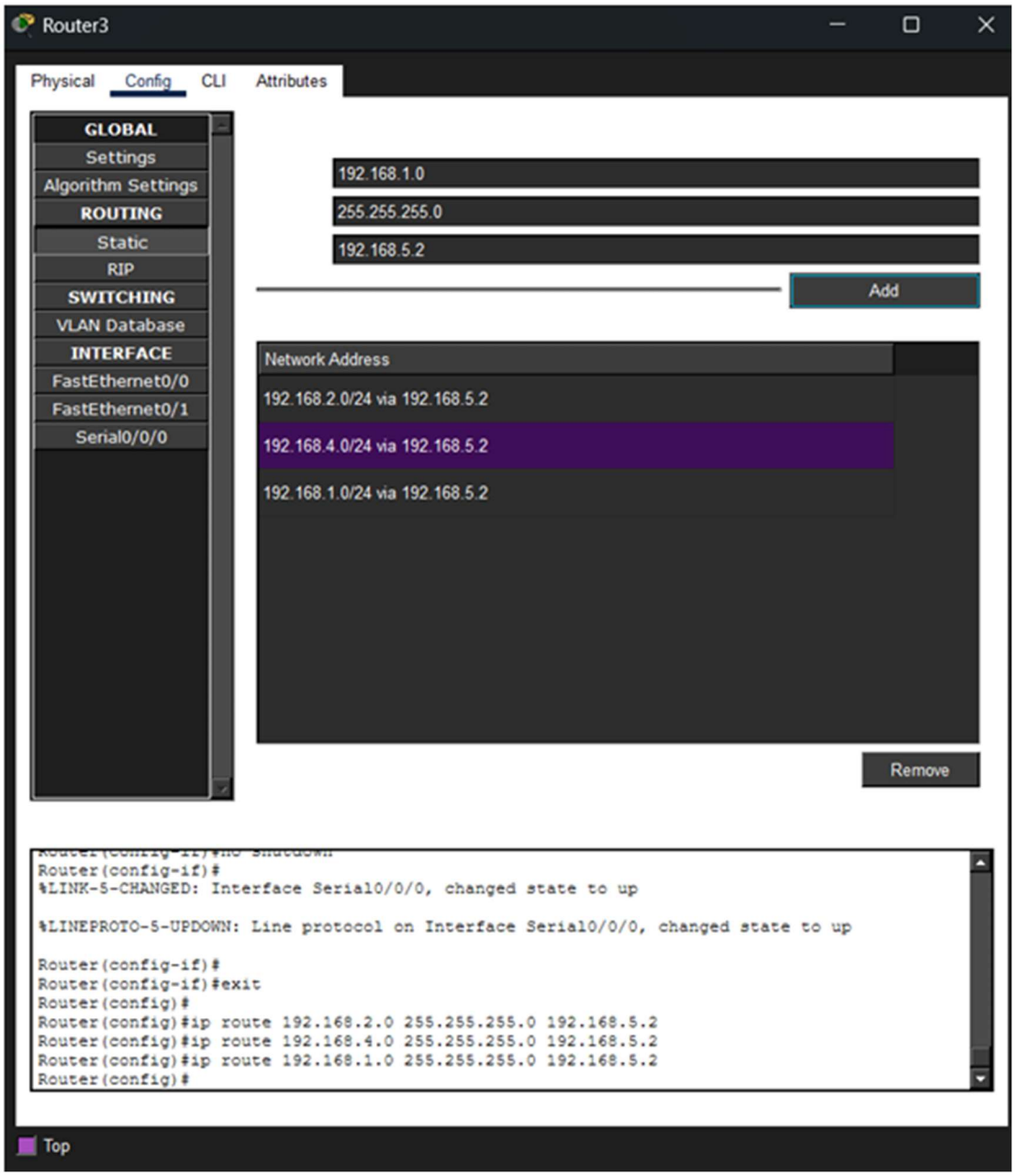

Figure 4 Router3 Routing Table

Figure 4 shows the static routes configured in Router3. Router3 requires routes to the left-side networks through Router1 using the next-hop address 192.168.5.2.

ip route 192.168.1.0 255.255.255.0 192.168.5.2
ip route 192.168.2.0 255.255.255.0 192.168.5.2
ip route 192.168.4.0 255.255.255.0 192.168.5.2

Observation Before Static Routing

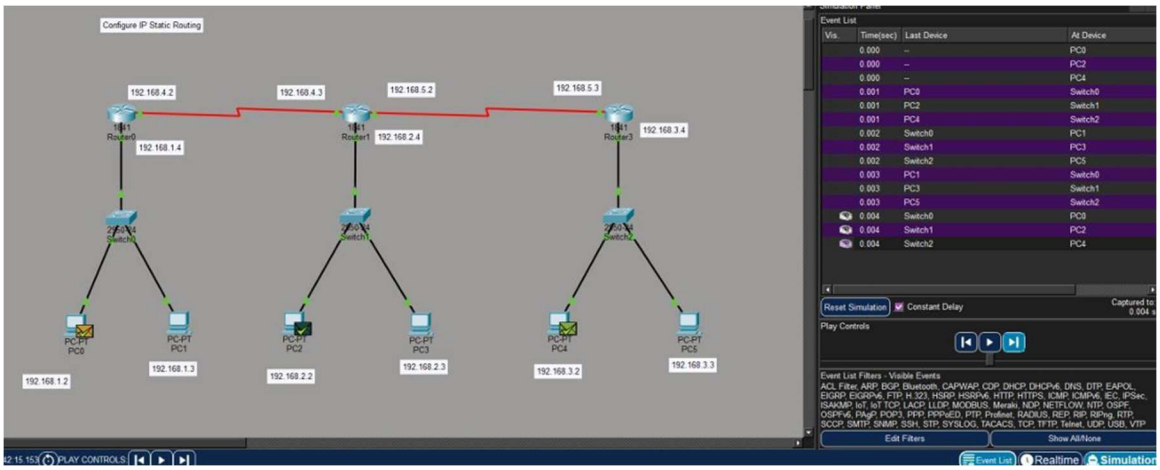

Figure 5 : Simulation before static routing.

Figure 5 shows that before configuring static routes, packets generated by PC0 can only travel to the switch and Router0. Router0 drops the packet because it does not know the route to the destination network.

Observation After Static Routing

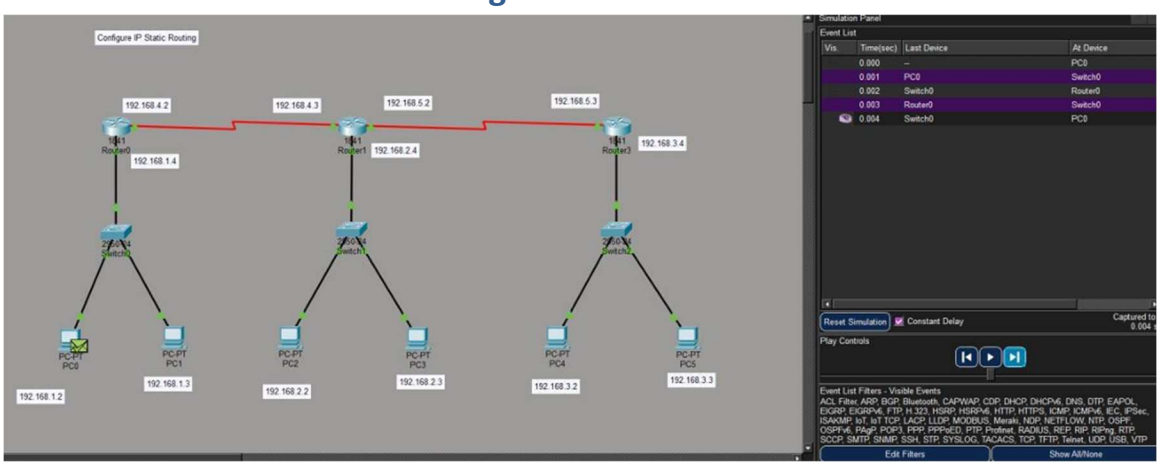

Figure 6 : Successful communication between different LANs

Figure 6 shows successful communication after configuring static routes. The packet now moves through multiple routers and reaches a device in another LAN.

The event list shows the complete path followed by the packet:
PC → Switch → Router0 → Router1 → Router3 → Switch → Destination PC

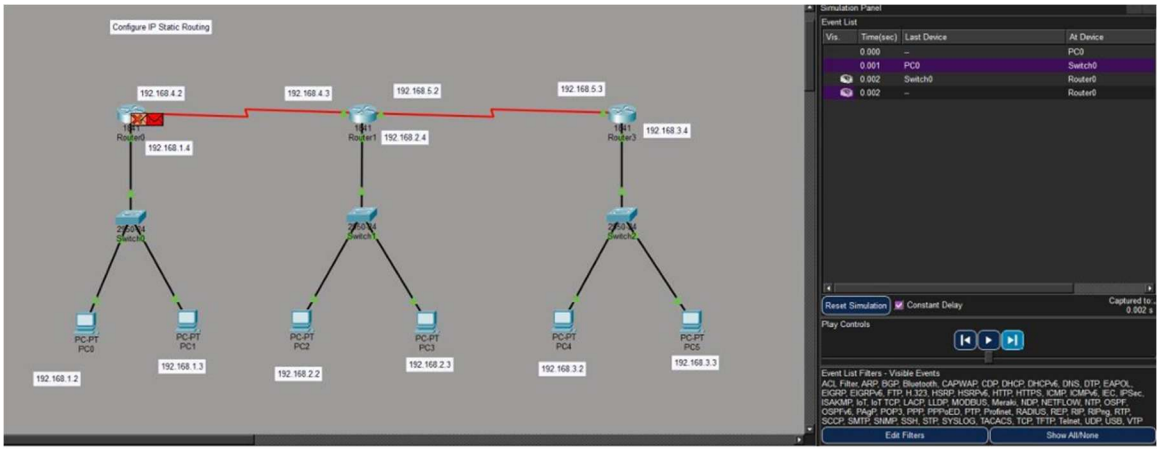

Figure 7

Figure 7 shows that when a packet is sent from a PC in one LAN to a PC in another LAN before configuring static routing, the packet reaches the local router but is not forwarded further. The router checks its routing table and finds no entry for the destination network, because it only knows its directly connected networks. Therefore, the router drops the packet and communication with the remote LAN fails. This demonstrates that routers require static routes to forward packets beyond their local networks.

Analysis

The screenshots show that before adding routes, routers only knew directly connected networks. Because of this, packets for unknown networks were dropped. After static routes were configured, every router had complete information about the remote networks. This enabled end-to-end communication between PCs belonging to different LANs.

Conclusion

The experiment successfully demonstrated static routing in Cisco Packet Tracer. Manual routes were added in all routers and communication between all three LANs became successful. Static routing is simple and useful for small networks, but it becomes difficult to manage when the network grows.